University of Nottingham
UK | CHINA | MALAYSIA

Literature Review- Computer Science dissertation

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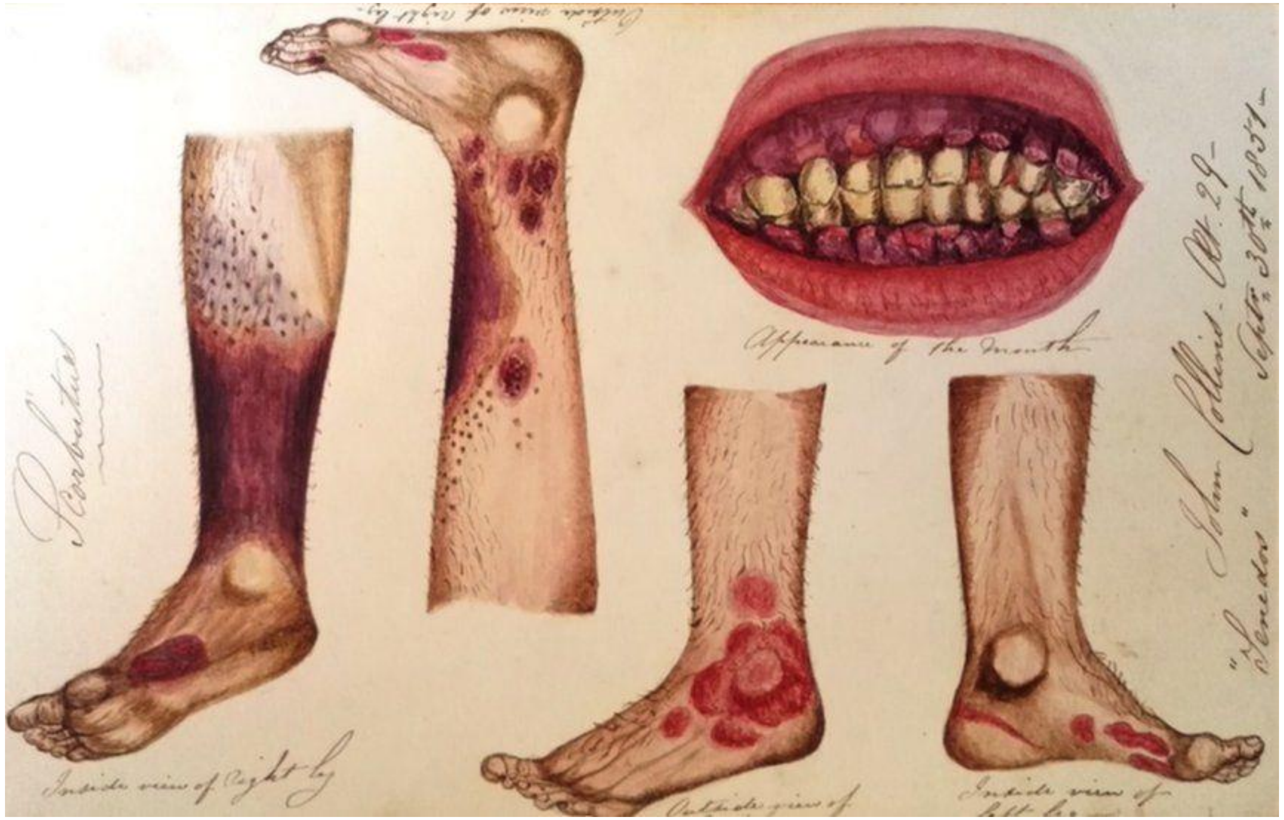
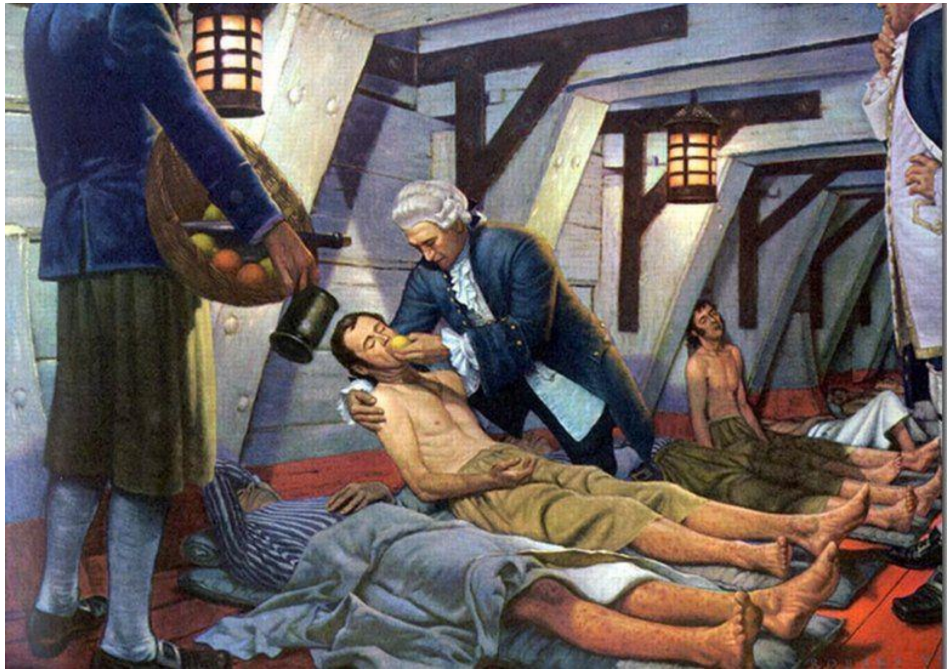
James Lind finds a cure for scurvy in 1747



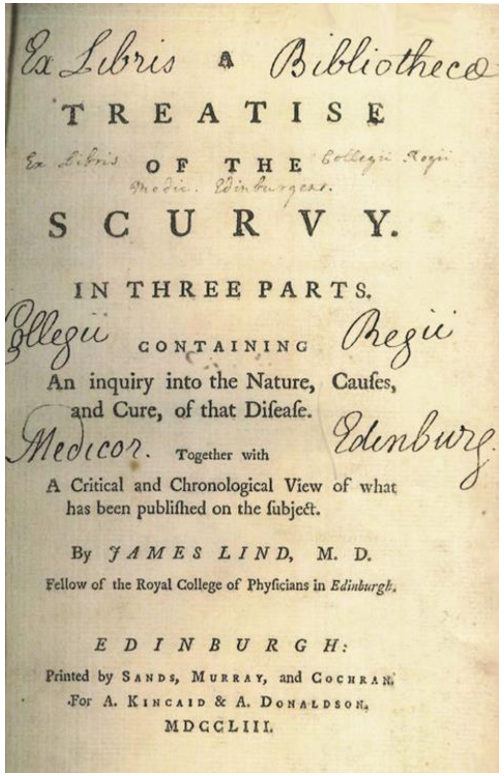
Vitamin C deficiency

Sailors die on long sea journeys



On 20 May 1747, James Lind took 12 'similar' patients with **scurvy** and divided them into six pairs. He carried out different treatments for each pair. Six days later, Lind reported:

The result of all my experiments was that ***oranges and lemons*** *were the most effectual remedies for this distemper at sea.*



Ex Libris Bibliotheca

TREATISE

Ex Libris OF THE Collegii Regii Medic. Edinburgensis.

SCURVY.

IN THREE PARTS.

Collegii CONTAINING Regii

An inquiry into the Nature, Causes, and Cure, of that Disease.

Medicor. Together with Edinburg

A Critical and Chronological View of what has been published on the subject.

By JAMES LIND, M. D.

Fellow of the Royal College of Physicians in Edinburgh.

EDINBURGH:

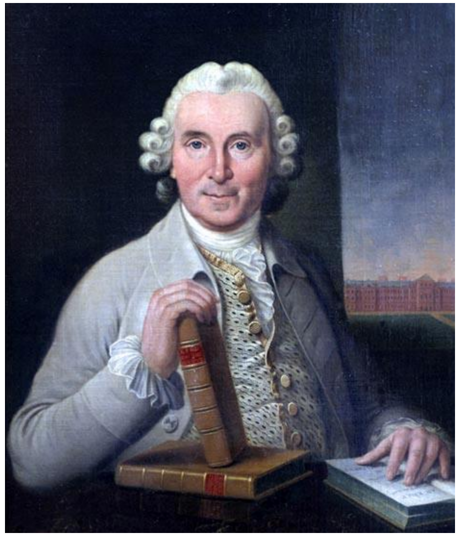
Printed by SANDS, MURRAY, and COCHRAN.

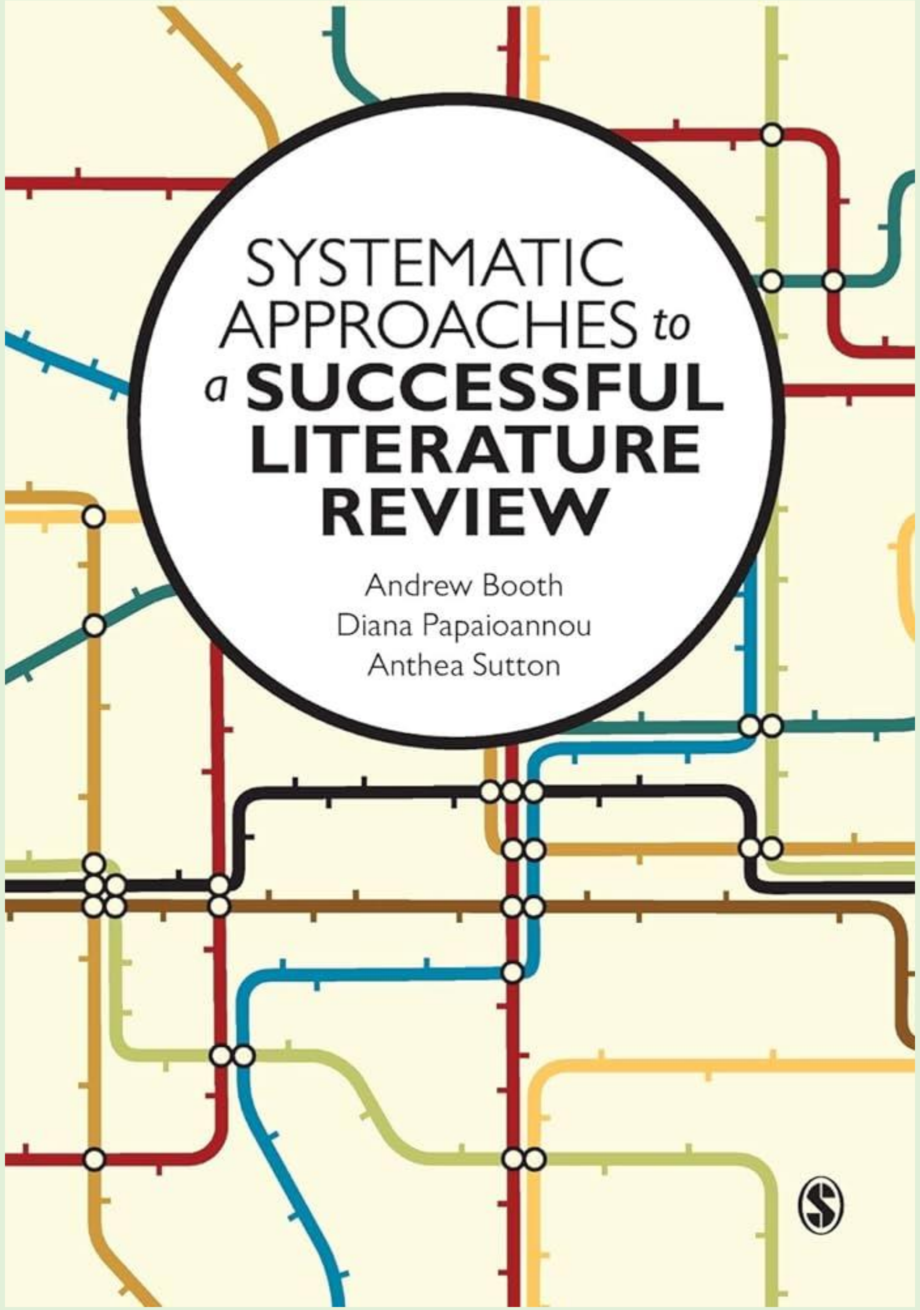
For A. KINCAID & A. DONALDSON,

MDCCLIII.

Lind acknowledged the need to **review existing literature systematically** and to discard 'weaker evidence': *As it is no easy matter to root out prejudices ... it became requisite to exhibit a full and impartial view of what had hitherto been published on the scurvy ... by which the sources of these mistakes may be detected. Indeed, before the subject could be set in a clear and proper light, it was necessary to remove a great deal of rubbish.*

The first systematic review / clinical trial




SYSTEMATIC APPROACHES to a SUCCESSFUL LITERATURE REVIEW
Andrew Booth
Diana Papaioannou
Anthea Sutton

Type of review	Purpose	Matt's requirements
Scoping review	• To find out how much literature exists • To find out the characteristics of the literature	• Overview of the field
Mapping Review	• Aims to **identify gaps** in the literature in order to conduct **further research**	• Overview of the field
Systematic review	• Aims to be **comprehensive** • Scope should be **well-defined** and **clearly stated** • Focus on the **highest quality evidence** available	• Critical review of specific area • State-of-the-art

Common reviewer errors (SALSA)

Search	• Does not identify the best sources to use in review literature related to the topic. Scope too limited • Does not report the search procedures that were used in the literature review.
Appraisal	• Relies on secondary sources rather than on primary sources. • **Uncritically** accepts another researcher's findings and interpretations as valid, rather than examining critically all aspects of the research design and analysis.
Synthesis	• Reports isolated statistical results rather than synthesising them • Reports isolated themes or categories rather than examining the relationships between them. • Does not consider contrary findings and alternative interpretations in synthesising quantitative or qualitative literature.
Analysis	• Does not clearly relate the findings of the literature review to the researcher's own study or the original review question.

Literature Reviews – comparison of 4

Name	Organization	Number of references & Citation type
A Virtual Reality and Wearable Device Technology System to Monitor Health Data and Deliver Phobia-motivated Experiences	**2 Background & Related Work** 2.1 VR Phobia Research 2.1.1 VR Exposure as a Safe Method to Assess Stress 2.1.2 Social Phobia Study 2.1.3 Smart Watch as a Stress-Measuring Tool 2.2 Industry Research 2.2.1 C2Care 2.2.2 Limbix 2.2.3 Qardio 2.3 Resources	23 citations 12 unique 11 repeated 3 Figures 6 pages
Comparing the Impact of Analogue versus Digital Control Panels upon Pilot's Mental Workload	**2 Literature Review/Related work** 2.1 MWL Related Assessment Techniques 2.1.1 Electroencephalogram & Event-Related Potentials 2.1.2 Heart Rate Variability 2.1.3 NASA Task Load Index 2.2 Information Processing Models 2.3 Existing Studies	54 citations 30 unique 24 repeated 8 pages
Machine Learning Methods for Single Cell Data Analysis using Single Cell RNA Expression	2. Literature Review 2.1 scRNA technology and data 2.2 PBMCs 2.3 Single-cell Data Analysis	14 citations None repeated 2 Figures 2 pages
Radar Human Activities Recognition Using Radar micro-Doppler Signature	**2 Literature review** 2.1 Doppler effect 2.2 Micro-Doppler signature 2.3 Representation of radar signal 2.3.1 Time-Doppler representation – Spectrogram 2.3.2 Range-Doppler representation – Range-Doppler map 2.4 Vision-based Gait Recognition 2.5 Previous Radar Human Activities Recognition (HAR) 2.5.1 Based on spectrograms (time-Doppler map) 2.5.2 Based on time-range map 2.5.3 Based on range-Doppler map	27 citations 20 unique 7 repeated 9 figures/ images 10 pages

Dissertation 1

Title	Citations, Figures, pages
A Virtual Reality and Wearable Device Technology System to Monitor Health Data and Deliver Phobia-motivated Experiences	23 citations 12 unique 11 repeated 3 Figures 6 pages

2 Background & Related Work

Dissertation 2

Title	Citations, Figures, pages
Comparing the Impact of Analogue versus Digital Control Panels upon Pilot's Mental Workload	54 citations 30 unique 24 repeated 8 figures 8 pages

Dissertation 3

Title	Citations, Figures, pages
Machine Learning Methods for Single Cell Data Analysis using Single Cell RNA Expression	14 citations None repeated 2 Figures 2 pages

Dissertation 4

Title	Citations, Figures, pages
Radar Human Activities Recognition Using Radar micro-Doppler Signature	27 citations 20 unique 7 repeated 9 figures/ images 10 pages

2 Literature review

Citations: Examples from student literature reviews

1	**End of sentence**	• Limbix is a product approved by the FDA for researching mental health issues using VR experiences [4]
2	**Within the sentence: single and multiple**	• VR therapy has been applied in many case studies, such as in fear of flying [19], nervousness in social situations, or social phobia [20], and acrophobia, a fear of heights [21] • C2Care [3] is one of the largest products on the market nowadays that helps treating phobias using VREs. • Currently, most researches [22], [33], [2] apply video gait data into deep learning algorithms for gait recognition. The Large Population Dataset established by Osaka University in 2012 [21] is the largest human gait database.

Citation style? Guide?
Consistency of use?
Citation mgt software?

3 **Start of sentences : Surnames / Family names and numbers**

- Qardio [5] is one of the best modern ways to monitor users' health activity.
- Lean et al. [36] have shown that the relationship between MWL and HRV is centred in the Autonomic nervous system.
- According to the conclusion drawn by Jiao et al. [37], the ratio LF/HF which represents the balance of sympathetic tone and vagal tone increases as MWL increases.
- According to Young and Stanton, both mental underload and overload will have a negative influence on human performance [12], it is important to keep the human MWL at an optimal level to minimize human errors and maximize system safety.
- According to O'Donnell and Roberts [15], most procedures for scientific evaluation can be classified into three main categories.
- In [26], Kim and Ling extracted the micro-Doppler signature and generated the spectrograms of echo signals.
- In [38], Shao et al. proposed three-layer DCNN to recognize six different human activities and obtained the performance of 95.2%
- proposed a HAR method that using DNN
- proposed CAE architecture to recognize 12 different human activities by using achieve a high performance of 94.2%.

Integral and non-integral citation methods. Stylistic choice/'voice' in writing

4	**Surnames numbers and brackets**	• More explicitly, several reviews identified that the alpha activity relates to the \default mode brain activity" (Laufs et al. [22], Jann et al. [23]) and it will decrease as MWL increases (Fink et al. [24], Brouwer et al. [25]), theta band can reflect the level of MWL and it will increase as MWL increases (Raghavachari et al. [26], Esposito et al. [27]). • Although several studies reported that there exists a relationship between the other 2 bands(δ & β) and human MWL (Brookings et al. [28]), the electrode sites that can observe the significant change of these bands (e.g., F3 & Fz shown in Fig 19) were not be used in this study (detailed in Section 4.3), hence only α band and θ band's activity were focused in this study • In other words, HRV will not sensitive enough to identify different MWL under different task demands if the demands changing is not large enough (Veltman and Gaillard [39]).

Mixture of two citation formats?

Tables and figures

1	**Start of sentence**	• The Figure 2.2 shows the micro-Doppler signature ofthe human body while walking. • Figure 2.3 (a) shows the range-Doppler map \video" which is the 3D representation of radar echoes.
2	**End of sentence**	• The amount of scRNA data is growing explosively (Figure 1).
3	**Middle of sentence**	• The bibliometric analysis shows that the number of scRNA-seq publications increases in an exponential rate (Figure 1b) and to date there are more than 14,000 publications referenced in Google Scholar, half of them published in the last two years • There is a total of seven different activities to be recognized (Figure 2.4 and Figure 2.5) and they proposed a customized Support Vector Machine (SVM) to realize activities recognition.
4	**End of sentence**	• For healthy individuals, the intervals between their successive heartbeats are different (shown in Fig 4)

Use of acronyms

1	**After term used for first time**	• Chronic lymphocytic leukemia (CLL) is one of the most common types of leukemia, • Peripheral blood mononuclear cells (PBMCs) • It can be derived by taking Short Time Fourier Transform (STFT) which is used to describe the frequency and time domain changes.
2	**Not given/ missing**	• 10x [3] is a popular SCT sequencing technology because it is affordable

Give full name and acronym first time. Then only acronym

Figures, Tables, websites

Two figures side by side

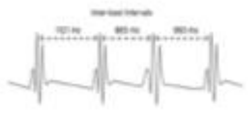


Figure 4: "irregular" heartbeats

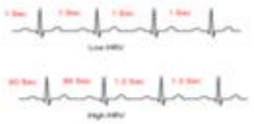


Figure 5: HRV example

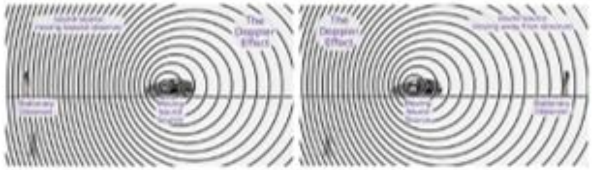


Figure 2.1: The image on the left shows the sound source is moving towards the observer, and the image on the right shows the sound source is moving away from the observer.

Single table

sub-band name	corresponding frequency
delta(δ)	$1 \sim 3$ Hz
theta(θ)	$4 \sim 7$ Hz
alpha(α)	$8 \sim 12$ Hz
beta(β)	$13 \sim 30$ Hz

Table 1: Corresponding Frequency of the sub-bands

Web page screen shot

Figure 2.3: C2Care VR Headsets Selection [3]

Website reference in Bibliography

[24] HTC. Htc vive vr headset. https://www.vive.com/us/product/. Accessed 16-May-2020.

[25] Oculus. Oculus rift s. https://www.oculus.com/rift-s/. Accessed 16-May-2020.

[26] Samsung. Samsung Gear VR. https://www.samsung.com/global/galaxy/gear-vr/. Accessed 24-March-2020.

[27] YouTube. About YouTube. https://www.youtube.com/intl/en-GB/about/. Accessed 18-December-2019.

Weaknesses in Knowledge and Understanding in 2 Literature reviews

Real2Sim based Intelligent Guided Vehicle (IGV) Routing for Container Port Terminals

No breakdown of the literature review sections

Subheadings within the literature review

Excessively detailed and attempting to cover too much.

Should select, prioritize, compare, contrast, synthesize, condense, link,

Advancements in Optimization Techniques for Industrial Systems and AGV Management

Enhancements in Genetic Algorithms and Particle Swarm Optimization

Integration of Genetic Algorithms in Complex System Configurations

Advanced Applications in Industrial and Energy Systems

Optimizing Operational Strategies in Container Terminals

Further Developments in Hybrid Optimization Techniques for Industrial Applications

Subheadings within the literature review

Early foundational work by KIM and TANCHOCO (1991)

Shortly after, SABUNCUOGLU and HOMMERTZHEIM (1992)

Further progress was made by Yan Zheng and Seo (

Capitalization errors

Building on these advancements, Fan et al. (2015)

The field saw additional advances with Zhuang et al. (2016),

The application of advanced optimization techniques began with Yiqing et al. (2007),

Subsequently, Lee and Hsiao (2012)

Sheikhalishahi et al. (2013) furthered this trend

Following these advancements, Sahoo et al. (2014)

Adding to these advances, Mousavi et al. (2017)

Following up, Yu and Xu (2014)

strial settings, Luo and Wu (2015)

Continuing this line of research, Jamrus et al. (2018)

In the specific context of container terminals, Yang et al. (2018) developed

Building on advances in hybrid optimization techniques, Jamrus et al. (2015)

- 15 citations are given in logical and chronological order.
- Brief summaries are given of the key findings for each paper

Their approach, focusing on optimizing scheduling through a novel time window graph concept, significantly improved route efficiency by managing vehicle reservations and scheduling conflicts

They highlighted the effectiveness of specific scheduling rules in minimizing job flow times in flexible manufacturing systems, adding valuable insi… scheduling processes.

Their method achieved significant imp…ments in computation times and …tion quality, further improving the efficien…

This meth… …tween different loops and balance the load within loops in a flexible manufacturing system, and was shown to be feasible and effective in real-world applications.

This method effectively managed static and dynamic obstacle… in uncertain environments, showcasing the flexib… …d broad applicability of hybrid optimizat…

This method impr… gene regulatory networks and demonstrated superior performance in network parameter estimation

Their method significantly improved computational efficiency and solution quality for complex NP-hard problems

Their approach effectively reduced makespan and the number of AGVs required while optimizing the use of AGVs' battery charge

Repetition of language and structure- every 2 sentences.

No details are selected for highlighting or analysis

All results seem to be highly, significantly, feasible, effective, optimal

Problems with literature review

- The literature review is a long list of studies- with a summary of a key finding for each. Each mini paragraph is 2 or 3 sentences, with a similar sentence structure
- It doesn't select specific details from a particular study and highlight their significance for the students particular study
- The studies are not compared and contrasted in any detail
- There doesn't seem to be much evaluation of specific studies or variables
- The limitations of the experimental set up, or the different conditions, in which each study was conducted are not referred to.
- All studies seem to report great results. No studies point to problems, failures, limitations, unexpected issues.

A better literature review may do the following:

Identifies key studies related to the students particular study. Prioritize the most important and least important

Shows how those studies relate to each other- making comparisons, contrasts, evaluations

Summarizes key ideas and synthesizes comparative data.

Points out limitations, problems, gaps in the literature

Avoids generalizations and is specific and detailed

Varies grammatical structures and avoids simple descriptive listing

Is every study a major success?

What limitations were in place for each study?

Variables, qualifications, simplifications, etc.

significantly improved route efficiency

achieved significant improvements

was shown to be feasible and effective

improved scalability

demonstrated superior performance

effectiveness of specific scheduling rules in minimizing job flow times

effectively managed static and dynamic obstacles

significantly improved computational efficiency

effectively reduced makespan

while optimizing the use of AGVs

Conformal Prediction| Decision Trees

2 Literature Review

The key to the integration of the decision tree algorithm and conformal prediction is the understanding of the structure of the decision tree algorithm and the theory of conformal prediction.

In the following sections, the traditional decision tree algorithm is presented first, followed by Iterative Bipartite 3 (ID3) and Classification and Regression Trees (CART). Finally, Conformal prediction, Inductive Conformal Machine (ICM), methods that can improve conformal prediction results Conformal Training (ConfTr), and a successful case of chimeric conformal prediction Directly Optimized Inductive Conformal Regression (DOICR) are introduced.

This looks more like an outline summary and description of these areas.

2.1 Decision Tree Algorithms

The Decision tree algorithm is a supervised learning algorithm. It is a typical classification algorithm to process data to generate the corresponding decision tree model. The decision tree algorithm can analyze the new unlabeled object. The decision tree model [9] is a hiera rules. It is used to recursively partition the feature space of a dataset into one-class subspa often used to find and extract features in large databases, which has an important impact on discrimination and prediction modeling.[10] At the same time, the decision tree algorithm can be used as a unit of the random forest algorithm [11] to participate in the prediction o the random forest algorithm. The random forest algorithm is a successful general classification and regression method.

References are given in number form.

This method u predictions of multiple random decision trees, and it performs well when the number of obs he number of variables. In the traditional decision tree, the value of information entropy about features is used to group training samples and set models. Assuming that a feature has m value levels, the information entropy of this feature is calculated as follows: entropy = −p1 ∗ log(p1) − p2 ∗ log(p2) − ... − pm ∗ log(pm) (1)

Specifically, the original training set $(x_1, y_1), ..., (x_l, y_l)$ of l examples is divided into a new training set $(x_1, y_1), ..., (x_m, y_m)$ with $m<l$ elements and a calibration set $(x_{m+1}, y_{m+1}), ..., (x_l, y_l)$ with $k := lm$ elements, where $x_i \in R^n$ are the attributes and $y_i \in R$ are the labels, $i = 1, ..., l$, and the attributes of a new example $x_{l+1} \in R^n$. The model trained on the new training set is used to make predictions on the calibration set. We associate an NCM between the predicted results and the corresponding target values of the calibration set. This nonconformity measure can be defined as

$$\alpha_i := |y_{m+i} - \hat{y}_{m+i}|, i = 1, ..., k, \tag{4}$$

where $\hat{y}_{m+i}$ is prediction value of point predictor. For each of the examples in the test set, we can define

$$\alpha_{k+1} := |y - \hat{y}_{l+1}|, \tag{5}$$

where $\hat{y}_{l+i}$ is the prediction for the new example. With Eq. 5 and α of the calibration set, we can get the potential target value for the new example. Let us define the p-value associated with the potential label y as

$$p(y) := \frac{\#\{i = 1, ..., k+1 : \alpha_i \geq \alpha_{k+1}\}}{k+1} \tag{6}$$

where #A is the number of elements in the set A. Given a confidence level $1 - \delta$, where $\delta>0$ is a small constant, the predictive region output by ICM is

$$y : p(y)>\delta \tag{7}$$

From the above formula, we can obtain the prediction interval for a new example by performing the following steps: Sort the *alpha* list after getting

References are given in number form.

Detailed mathematical notation

Should this be in a different part of the dissertation?

The structure of the model and its ability to generalize can also affect the results. Therefore, machine learning models are often sensitive to external factors. It's particularly important for getting the accuracy of the current prediction and the corresponding target value range. Conformal prediction (CP) [14] is a method of using models to obtain prediction accuracy, which uses past experience to determine the precise confidence level of a new prediction. Specifically, a point prediction model makes a prediction $\hat{y}$. For this prediction $\hat{y}$, conformal prediction can generate a 95% prediction region (confidence interval)-a set $\Gamma^{0.05}$, which contains actual observed value with probability at least 95%. This probability is also called the confidence level. Confidence intervals always contain $\hat{y}$, but not necessarily the correct ones.

A sample set is commutative if it has equal probabilities under different orderings. Under exchangeability, the assertion of confidence 95% is valid, independent of the probability distribution followed by the samples and the nonconformity measure used to construct conformal prediction regions.[14] However, the effectiveness of conformal prediction will depend on the probability distribution and the nonconformity measure.

Conformal prediction can be used in many machine learning methods, including Bayesian prediction, vector machines, neural networks, and decision trees. To generate a prediction region, we need the nonconformity measure (NCM) from the point prediction. NCM measures how different the current example is from the previous examples. After obtaining the NCM, the conformal prediction algorithm can convert it into the corresponding prediction

Long passages outlining or defining various methods (Conformal Prediction)

Reference 14 used twice in these 3 paragraphs

Should this be in a different part of the dissertation?

inflexible. The loss function is used at training time to minimize the entropy of the model, and the model is used at testing time to obtain a confidence set with a specified confidence level. The whole model training and testing process cannot reduce the resulting prediction interval. Bellotti (2021) [16] solves this problem by learning a set predictor $C(X)$ through thresholding logits. When classes with logits exceed 1, then put them in $C(X)$ and train to reduce the inefficiency while targeting coverage 1 - α.

9 references in total.

Mixed referencing methods: Belotti(2021) + [16]

Some features of literature reviews

Structure, Roadmap, Conclusion	Functional Tasks 1	Functional tasks 2	Grammar, Vocabulary
Purpose statements and Roadmaps	Terminology and concepts- definitions, short quotations	Stating the results of studies / individual findings	Comparing and contrasting
Justification for this study and approach	Definitions- extended	Summary of a study- simple / extended	Descriptive writing- Present tense verbs
Reference to recent studies and findings	Categorizing information	Synthesizing information from different studies	Text flow- Linking nouns and pronouns
Restricting the scope of analysis		Giving Reasons	Writing Style- Errors in formality
Conclusion of Lit Review		Hedging and qualifying statements	Vocabulary choice- dictionary use and synonyms
			Grammar Problems- overcomplicated sentences comma splice

MANCHESTER 1824
The University of Manchester

Academic Phrasebank

Search

Introducing work | Referring to sources | Describing methods | Reporting results | Discussing findings | Writing conclusions

Academic Phrasebank

GENERAL LANGUAGE FUNCTIONS

- Being cautious
- Being critical
- Classifying and listing
- Compare and contrast
- Defining terms
- Describing trends
- Describing quantities
- Explaining causality
- Giving examples
- Signalling transition
- Writing about the past

Home page

The Academic Phrasebank is a general resource for academic writers. It aims to provide you with examples of some of the phraseological 'nuts and bolts' of writing organised according to the main sections of a research paper or dissertation (see the top menu). Other phrases are listed under the more general communicative functions of academic writing (see the menu on the left). The resource should be particularly useful for writers who need to report their research work. The phrases, and the headings under which they are listed, can be used simply to assist you in thinking about the content and organisation of your own writing, or the phrases can be incorporated into your writing where this is appropriate. In most cases, a certain amount of creativity and adaptation will be necessary when a phrase is used. The items in the Academic Phrasebank are mostly content neutral and generic in nature; in using them, therefore, you are not stealing other people's ideas and this does not constitute plagiarism. For some of the entries, specific content words have been included for illustrative purposes, and these should be substituted when the phrases are used. The resource was designed primarily for academic and scientific writers who are non-native speakers of English. However, native speaker writers may still find much of the material helpful. In fact, recent data suggest that the majority of users are native speakers of English. More about **Academic Phrasebank**.

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Are you a postgraduate student with a specific learning difficulty? We are conducting research into how Academic Phrasebank can be useful for postgraduate students who have specific learning difficulties. Would you like to assist with this project by taking part in an online

https://www.phrasebank.manchester.ac.uk/

Structure,
Roadmap,
Conclusion

1. Purpose Statements and Roadmaps

- This chapter outlines major studies on VR exposure as a way to treat phobias, and wearable technology as a valid method to record stress levels. This chapter also includes the latest products on the market related to the VR healthcare industry

- This section describes the potential advantages of VR exposure on patients and the risks that could be caused by it. It also argues why wearable technology is becoming one of the effective tools to collect personal health data from the human body

- This section describes a social anxiety study [20]

- This section describes the products in detail and their potential impact on the market and this project.

- Hence, this section will first describe three psychophysiological measures and the reason why using them in this project. Then two information-processing models will be introduced which can provide a clear description of why and when the human's MWL will change. Finally, current studies related to the pilot's MWL will be discussed

- This chapter will include an overview of the studies in this field, focusing on attack-agnostic methods for estimating neural network robustness in both direct and indirect measurement approaches. Meanwhile, a retrospective review and evaluation of primary research methods in this area is presented to enhance understanding of this project.

1. Purpose Statements and Roadmaps

- In the following sections, the Doppler effect, micro-Doppler effect, and signatures are introduced briefly first, followed by the introduction of the representation schemes spectrogram and range-Doppler map. Then, the vision-based gait recognition and the previous radar human activities recognition are discussed in the last sections.
- As stated in Chapter 1, there are multiple applications on the market
- As discussed in Section 3.1.2, CLEVER are supported by a rich statistical theory and implemented easily in practice, but this method is prone to inaccuracies.
- All the above methods are essentially estimating Lipschitz constant.

2. Restricting the scope of analysis

- Only Subjective measures and Physiological measures will be discussed in this part.
- Although several studies reported that there exists a relationship between the other 2 bands(δ & β) and human MWL (Brookings et al. [28]), the electrode sites that can observe the significant change of these bands (e.g., F3 & Fz shown in Fig 19) were not be used in this study (detailed in Section 4.3), **hence** only α band and θ band's activity were focused in this study

3. Reference to recent studies and findings

- A recent study about human emotions indicates that a person’s body temperature, blood pressure, and heart rates are changing, rising in all cases, when a person is facing a stressful situation [22].

- A study on the validation of the Apple Smart Watch shows that 90% of all the heartbeats that were recorded during the testing stages were valid and correct, showing the same results as industrial heart rate monitors [23].

- As mentioned before, EEG data can be decomposed into several frequencies for analysis. So et al. [20] found that the relative energy of delta, theta, alpha and beta rhythms can reflect the change of MWL of the tested person and Demos and John’s research detected the corresponding frequency of these sub-bands [21] (shown in Table 1).

- Nowadays, radar has been applied to gait recognition and human activity recognition as an ideal tool. There have been researches on radar human activities recognition, and the detailed description will be presented in the next section

- The studies presented here indicate that VR exposure is an effective method to treat phobias and possibly replace expensive and time-consuming in-vivo therapy

- The results indicate that wearable technology can compete with industrial heart rate monitors on the open market as a tool to measure heart conditions.

4. Justification for your study / scope

- However, there are not many studies related to how different CDIs impact pilot’s MWL during flight.

- Based on the theory that HSI has a significant impact on MWL [45], it is deserving to find out the relationship between different layouts of the CDI and MWL of pilots which has not been studied in depth

- A key issue in research on measuring robustness is that the robustness evaluation result is often intertwined with the attack algorithms. However, the attack algorithms easily overestimate the amount of perturbation needed to confuse the target neural network since the attack algorithms are not optimal and comprehensive. In other words, the methodology based on the attack algorithms gives the upper bound βU, but the lower bound βL is needed for safety guarantees [18] which ensures the neural network is robust to any perturbation with kδk(pd) ≤βL.

Concluding the Literature Review

- Chapter 2 has summarized the background information from the related work and available products in the VR healthcare industry today. The next chapter introduces the software requirements specifications for the project and the design of the system

- There have been researches on radar human activities recognition, and the detailed description will be presented in the next section.

- Gradient masking [18] is one strong piece of evidences which proves CLEVER overestimates the adversarial perturbation size. In what follows, we also demonstrate that these inaccuracies do indeed affect the result returned by Algorithm 1.

Functional

Tasks 1

Terminology and concepts- definitions, short quotations

- Although the concept of MWL has been studied for nearly 40 years, its definition is still controversial [10].

- Ecological validity can be explained as the degree of whether the laboratory result can be generalized to the natural world [29], for example, the regular operational environments like the cockpit of an aircraft.

- Cognition can be defined as the mental process of getting knowledge and understanding, these processes are called cognitive processes which include \thinking, knowing, remembering, judging, and problem-solving“

- More explicitly, several reviews identified that the alpha activity relates to the “default mode brain activity" (Laufs et al. [22], Jann et al. [23]) and it will decrease as MWL increases (Fink et al. [24], Brouwer et al. [25]), theta band can reflect the level of MWL and it will increase as MWL increases (Raghavachari et al. [26], Esposito et al. [27])

Definitions- extended

Although the concept of mental workload (MWL) has been put forward and studied for nearly 40 years, its definition is still controversial [10]. Intuitively, the construct of MWL can be described as the amount of necessary mental work consumed by a certain task over a given period of time [11], which is affected by multiple extra factors at the same time, such as physical demands [10]. In general, MWL is a key criterion for evaluating operator and system performance [11], and designers can design a more suitable system for their users by estimating their MWL. Furthermore, based on the theory that individuals' MWL is more likely to increase when they operate a complex system [3], the potentiality of mental overload is higher these days because modern systems have become increasingly complex as a result of technological development [12].

Categorizing information

- According to O’Donnell and Roberts [15], most procedures for scientific evaluation can be classified into three main categories. The first one is Subjective measures,   … Secondly, Physiological measures represents… The last category is Performance-based measures.
- Wickens 4D Multiple Resources Model [44] describes three stages during information processing.
- There are two main approaches: Virtual Reality Exposure Therapy (VRET) and in-vivo therapy [18].

Functional

Tasks 2

1. Stating the results of studies / individual findings

- It was confirmed that VR therapy lowered participants' stress levels and reduced anxiety [19, 20, 21].
- Lean et al. [36] have shown that the relationship between MWL and HRV is centred in the Autonomic nervous system.
- The description given by Wilson et al. [43] (Fig 7) shows that human's capacity of processing information is limited, therefore when the spare capacity getting close to zero, the performance of the primary task will start to decline. The spare capacity can be measured by a technology named \secondary task" [43].
- Wei et al. [50] found that the accuracy of the judgement of situation awareness (SA) is different when pilots are dealing with information on different CDIs.
- For example, 93% of the people who were afraid of flying and went through VRET, flew on airplanes after several sessions without any fear [19].

2. Summary of a study- simple / extended

- Wei et al.'s [8] study concentrated on the performance under increasingly MWL on a specific aircraft on a flight simulator. Di et al. [48] used a flight simulation game for study and found that during the whole flight process, pilots' MWL is different at different stages which has a different impact on pilots' performance.

- For example, Morris and Leung's [6] study focused on how different did pilots perform when they tried to finish a desktop, computer-based exercise that simulated a set of flight deck activities on a particular type of aircraft under different MWL. The participants were divided into 3 MWL groups (low, medium, high) and were required to finish a task that contained a different level of cognitive demands. The result shows that the participants' primary task's performance decreased significantly as the cognitive demands increased.

- Scanman and Virmaux [12] have argued that exact Lipschitz computation is NP-hard, since this problem can be reduced to maximize a convex quadratic function on the hypercube which is a known NP-hard problem. Then, they propose a new algorithm—SeqLip which exploiting the relationships between singular vectors of consecutive layers to estimate Lipschitz constant:

3. Synthesizing information from different studies

- HR varies greatly under different conditions. For instance, De Rivecourt et al.'s [31] simulated flight study observed that the HR data would increase as the task's demand increases. However, Aasman et al. [32] found that HR has been shown not to be as sensitive to variations in MWL in \laboratory studies that use mental loading tasks.

- Compared with HR, Fallahi et al.'s [33] data showed that HRV can be affected as conditions changed from low to high traffic density in a traffic monitoring job. The same change was observed by Delaney and Brodie during a Stroop test [34]. In addition, the pilot task conducted by Jaiswal et al. [35] proved that using HRV as an assessment technique is workable

- Many existing studies have identified that a human-system interface (HSI) can have a significant impact on an operator's MWL. According to the data provided by Piechulla et al. [45], an adaptive HSI is helpful to reduce driver's MWL during driving. Hancock and Chignell [46] proved that MWL is dynamic when using different HSI. The same conclusion has been drawn by Jou et al. [47] when they tried to find out how the advanced nuclear power plants HSI automation affect the operator's MWL

3. Synthesizing information- identifying a problem/limitation

- In 2017, Hein & Andriushchenko [19] provide a guaranteed lower bound for the robustness of one hidden layer MLP by using a local Lipschitz continuous condition. But it is challenging for neural networks with more than one hidden layer to derive their theory. Weng et al. [10] brought an extended formal robustness guarantees for a classifier based on Hein & Andriushchenko's theory. Their extended theory is universal, since Lipschitz continuity of the classification function is the only required assumption. Based on their work, a theoretical robustness guarantee on the lower bound of minimum distortion βL for untargeted attack is given in Theorem 1

Giving Reasons

- In fact, EEG has been successfully used for several pilot's MWL related experiments in the past 30 years [19]. Hence EEG has been chosen as an assessment technique in this project. Hence, HRV has been chosen as the measure in this project.

- According to the conclusion drawn by Jiao et al. [37], the ratio LF/HF which represents the balance of sympathetic tone and vagal tone increases as MWL increases. The same trend was observed by Murata and Atsuo [38]. Therefore, the LF/HF ratio of the HRV data was focused on in this study

- For the target recognition, It is believed that the micro-Doppler signatures are different among different activities, different genders, or different individuals [42]. Therefore, m-Ds is a unique signature that can be used to realize radar human activities recognition.

- Moreover, as the difficulty of computing exact Lipschitz constant value, a useful estimate should be a tight upper bound of the Lipschitz constant. The reason is that, a lower bound of the Lipschitz constant would result in false negatives

- However, it have been proved that estimation of exact Lipschitz constant is a NP-hard problem [12]. Thereby, the methods discussed above are not optimized and have their pros and cons.

Hedging and qualifying statements

- One issue that might influence the research result by using HRV is that the HRV data might be more suitable to differentiate the MWL changing between the rest period and the task execution in aviation research (Wilson and Glenn [5]).
- In other words, HRV will not be sensitive enough to identify different MWL under different task demands if the demands changing is not large enough (Veltman and Gaillard [39]).
- Therefore the HRV data analysis might not be able to indicate a clear conclusion about the changing of MWL. Despite the “insensitive" problem, the sensors used to monitor the wearer’s HRV will cause issues related to ecological validity (mentioned in Section 2.1.1).
- Extra sensors placed on the participants might influence their HRV and the reliability of the HRV data might interfere

Grammar and Vocabulary

Comparing and contrasting

- Furthermore, similar to C2Phobia, Limbix does not collect any personal data. The Limbix kit is used only to play the VREs as video files.
- VRET is different from in-vivo therapy, where the patients or users would be accompanied by a psychologist or a related-field healthcare professional.
- One of the significant applications of SCT data analysis is to complete Human Cell Atlas so that cells extracted from new samples can be compared with a reference
- The learning and classification processes of the naïve Bayes method are extremely fast compared to other ML methods
- Unlike optical instruments, radar echo data often contains noise, such as external cosmic noise and internal thermal noise.
- Compared with recognition based on videos or images captured by optical equipment, radar recognition still has good performance in adverse weather or through walls.
- The experiment mainly compares the performance of RGEI, spatial-temporal vectors, single RDm, and mean Doppler images

Descriptive writing- Present tense verbs

- C2Phobia is one of the largest Food and Drug Administration (FDA) approved systems on the market, developing VR applications in the healthcare industry to treat disorders, addictions, and phobias. However, C2Phobia does not collect any personal data or notify the healthcare professionals about the sessions that the user has taken. C2Phobia works only as a VRE media player for displaying phobias, but it does not evaluate the impact of the VRE. Nevertheless, C2Phobia is still one of the first and largest products on the market that shows the combined technology working between the healthcare and VRindustries.

- Limbix is a product approved by the FDA for researching mental health issues using VR experiences [4].

- Qardio [5] is one of the best modern ways to monitor users' health activity. Qardio specializes in measuring heart rate,

Text flow- Linking nouns and pronouns

- Qardio [5] is one of the best modern ways to monitor users' health activity. Qardio specializes in measuring heart rate, electrocardiogram (ECG), respiratory rate and skin temperature. Qardio has different devices, such as QardioArm, QardioBase and QardioCore, each one for a different use. All of them are focused on checking the user's heart activity and reporting it to the application on the phone. Qardio works for both iPhone Operating System (iOS) and Android, including wearable technology (Figure 2.5). It comes with an application for mobile phones with various features, including storing, sharing and evaluating the data of the user. Qardio (restate main noun) also has a special bundle for healthcare professionals with extra features which are approved by many administrations all over the world.

- Qardio, Limbix, and C2Care are the three main products that are present in the healthcare market nowadays. They are efficient in what they do, but they are lacking some functionality, such as the collection of personal data, live images, wearable technology or VR headsets for mobile phones.

I or We or ?

- To assess generalization, we performed ten runs of the system and compared the results of all individual cycles within each run.

- Additionally, we plan to deploy this system on the PBMC data sets from disease samples to explore the potential of distinguishing healthy cells and diseased cells. We also plan to deploy our system for the classification of other cell types with available SCT data sets.

- Our preliminary estimation is that decision trees, random forests and KNNs are not suitable multi-class predictors

- In what follows, we also demonstrate that these inaccuracies do indeed affect the result returned by Algorithm 1

Writing Style- Errors in formality

- The Doppler effect refers to the frequency change of a wave caused by the object moving towards or away from the source [12] Here is a common example. As the observer watches a train pass by, the whistle becomes louder but thinner in pitch as the train approaches the observer, while as the train moves away from the observer, the whistle becomes lighter but more intense (Figure 2.1).
- There are lots of representation schemes of time-varying micro-Doppler signatures, such as spectrogram and cepstrogram.
- Continuous-wave radar (CW) has become a hot choice due to its low power consumption, strong anti-interference ability, long exposure time to capture more information, and other advantages [34].
- For example, the failed spectrogram I got at the beginning of the Chapter 2.

Vocabulary choice- dictionary use and synonyms

- Based on the theory that MWL is susceptible to physiological measures [13], it is reliable to evaluate MWL by related assessment techniques

- Currently, most researches [22], [33], [2] apply video gait data into deep learning algorithms for gait recognition.

- Since scRNA-seq was first deployed in 2009 [18], various flavors of scRNA technologies have been developed and implemented.

Grammar Problems- comma splice

- According to Young and Stanton, both mental underload and overload will have a negative influence on human performance [12], it is important to keep the human MWL at an optimal level to minimize human errors and maximize system safety.

- In general, MWL is a key criterion for evaluating operator and system performance [11], designers can design a more suitable system for their users by estimating their MWL.

Grammar Problems- overcomplicated sentences

- According to Charles and Nixon [13], EEG data can be decomposed in a needed frequency range, which makes it more flexible for different researches, for example, Ogilvie et al. [17] used Fast Fourier Transform (FFT) method to decompose the EEG data into 1 Hz to 30 Hz for analysing the difference of brain behaviour between sleep and awake state whereas Adeli et al. [18] used Wavelet Transform to pick the 3 Hz EEG data to analyse an epileptic patient’s brain activity. (81 words)

- 10x [3] is a popular SCT sequencing technology because it is affordable (largescale measurement can be done at low cost), more than 100,000 cells can be profiled in a single study, typically within 24 hours, and the methodology has strict Standard Operating Procedures (SOP) that results in reproducibility of measurements (50 words)

Referencing errors and mistakes

- Based on the binary silhouette sequence, J. Han proposed Gait Energy Imaging to extract features [15]

- Gait recognition is a biometric recognition technology that recognizes the dynamic feature of the human body. Compared with other biometric features, such as fingerprints and iris, gait recognition has the advantage of being capable of remote detection without subjective cooperation of the subject. Inspired by the data processing method of vision-based gait recognition GEI, this dissertation proposes the Radar Gait Energy Imaging method, and applies it to human activity recognition, and has achieved satisfactory results

- Peripheral blood mononuclear cells (PBMCs) are blood cells identified by the presence of a single round nucleus. Human PBMCs include five main cell types: T cells (TC), B cells (BC), NK cells (NK), monocytes (MC), and dendritic cells (DC) [20]. The frequencies of five major cell types in PBMCs vary across different individuals of various health conditions and other biological specificities. Through comparing multiple antibody catalog estimates, a rough consensus is that T cells make 40–70%, B cells make 5–15%, NK cells make 5–10%, monocytes make 10–30% and dendritic cells make 1–2% of PBMC in humans. PBMCs are commonly used in medical blood testing and can be simply isolated from mixed blood samples. Human healthy PBMCs data were selected as a case study in this project.

Lecture evaluation and Writing Lab advising

What does ‘**systematic**’ look like?

- Clear specification of planned **review methods/protocol**
- A clearly **focused question**
- Clear, explicit **criteria** for inclusion and exclusion
- Documentation of **search process**: sources and strategies
- Use of tables and boxes to make **methods** explicit
- Use of tables to summarise **study characteristics**
- An <u>explicit mechanism</u> to handle **quality assessment**
- Exploration of **assumptions, limitations** and areas of uncertainty
- Use of tables and graphics to support **interpretation of data**
- Appendices including <u>search strategies, sample data extraction and quality assessment tools</u>